MEME

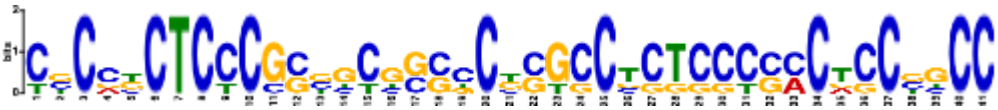
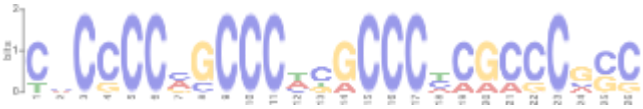
Multiple Em for Motif Elicitation

For further information on how to interpret these results please access https://meme-suite.org/meme/doc/meme.html.
To get a copy of the MEME software please access https://meme-suite.org.

If you use MEME in your research, please cite the following paper:
Timothy L. Bailey and Charles Elkan, "Fitting a mixture model by expectation maximization to discover motifs in biopolymers", *Proceedings of the Second International Conference on Intelligent Systems for Molecular Biology*, pp. 28-36, AAAI Press, Menlo Park, California, 1994. [full text]

Discovered Motifs | Motif Locations | Inputs & Settings | Program Information | Results in Text Format | Results in XML Format

Discovered Motifs

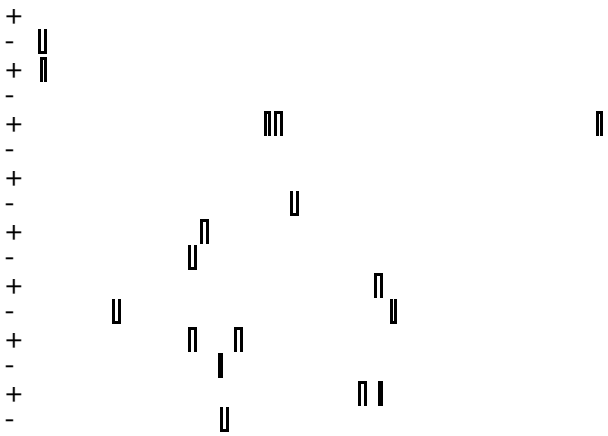
		Logo	E-value	Site
1.	+ -		3.6e-003	
2.	+ -		2.1e-001	
3.	+ -		3.2e+000	

Stopped because requested number of motifs (3) found.

Motif Locations

Only Motif Sites | Motif Sites+Scanned Sites | All Sequences | Download PDF | Download SVG

	Name	*p*-value	Motif Locations
1.	chr12:9435427-9436417(+)	9.03e-9	
2.	chr12:3841458-3842476(+)	3.65e-7	
3.	chr12:58334036-58336933(+)	2.39e-28	
4.	chr12:60492541-60494268(+)	6.70e-8	
6.	chr12:56914843-56918281(+)	9.92e-20	
8.	chr12:54426167-54428690(+)	2.24e-27	
9.	chr12:54088501-54090917(+)	3.10e-29	
10.	chr12:123754071-123755903(+)	5.02e-34	



Inputs & Settings

Sequences

Role	Source	Alphabet	Sequence Count	Total Size
Primary Sequences	data_s1_H3K4me3.bed.fa	DNA	10	18967

Background Model

Source: built from the (primary) sequences

Order: 0

Name	Freq.	Bg.				Bg.	Freq.	Name
Adenine	0.253	0.253	A	~	T	0.253	0.253	Thymine
Cytosine	0.247	0.247	C	~	G	0.247	0.247	Guanine

Other Settings

Motif Site Distribution	ZOOPS: Zero or one site per sequence
Objective Function	E-value of product of p-values
Starting Point Function	E-value of product of p-values
Site Strand Handling	Sites may be on either strand
Maximum Number of Motifs	3
Motif E-value Threshold	no limit
Minimum Motif Width	6
Maximum Motif Width	50
Minimum Sites per Motif	2
Maximum Sites per Motif	10

Show Advanced Settings

MEME version
5.5.8 (Release date: Thu May 15 15:01:46 2025 -0700)

Reference
Timothy L. Bailey and Charles Elkan, "Fitting a mixture model by expectation maximization to discover motifs in biopolymers", *Proceedings of the Second International Conference on Intelligent Systems for Molecular Biology*, pp. 28-36, AAAI Press, Menlo Park, California, 1994. [full text]

Command line

```
meme data_s1_H3K4me3.bed.fa -dna -oc . -nostatus -time 14400 -mod zoops -nmotifs 3 -minw 6 -maxw 50 -objfun classic -revcomp -markov_order 0
```